

# CSS Animations & Transitions

*Make Your Webpage Come ALIVE!*



Beginner



Code Editor + Browser



**Objective:** Understand CSS transitions and create smooth hover effects



*Keep coding — keep creating magic!*



# What Is Animation?

Animation makes things MOVE, CHANGE, or come ALIVE — guiding attention and making web experiences feel polished!



## Swiggy Loader

Bike rides across screen while food is prepared



## PhonePe Tick

Green tick appears after a successful payment



## Google Doodle

Letters bounce or transform on special days



## Diwali Diya

Flame flickers on festival websites



## Flipkart Buttons

Glow or grow when you hover over them

*Without animation: flat & jarring → With animation: alive & professional ✨*

# CSS — The Magic Wand

**CSS (Cascading Style Sheets)** controls colours, sizes, fonts, spacing — and YES, animations! A few lines of code tell the browser how to move, change colour, grow, shrink, or spin.

## Two CSS Animation Tools

### CSS Transition

 What: Smooth change between two states

 When: On hover, click, or focus

### CSS @keyframes

 What: Continuous or complex movement

 When: Looping effects, loading spinners

★ Session focuses on CSS Transition. @keyframes comes in Session 2!

# CSS Transition: The Slow-Change Tool

## Real-Life Analogies



### Traffic Light

Red → Yellow → Green — smooth, ordered transitions



### Sunrise Sky

Black → purples → pinks → oranges — a slow beautiful blend



### Flipkart Button

Colour fades from white to orange over a fraction of a second

## The Transition Syntax

```
/* Basic syntax */  
transition: background-color 1s;  
  
/* Multiple properties */  
transition: background-color 1s, transform 0.5s;  
  
/* Transition ALL properties */  
transition: all 0.8s;
```

# The 3 Parts of a CSS Transition 1 2 3 4

1

## The Selector

```
.box { }
```

Targets which element gets the magic — a div, button, image, or any HTML element.

2

## The Transition Property

```
transition: background-color 1s;
```

The magic line! What to change + how long. Goes in the NORMAL style, NOT in :hover.

3

## The Hover Trigger

```
.box:hover { }
```

Fires when the user's mouse moves over the element. Like a motion-sensor light switch!



# Without vs. With Transition

## Without Transition

```
.box {  
  background-color: orange;  
}  
  
.box:hover {  
  background-color: green;  
}
```

 Color flips instantly — jarring and abrupt!

VS

## With Transition

```
.box {  
  background-color: orange;  
  transition: background-color 1s;  
}  
  
.box:hover {  
  background-color: green;  
}
```

 Color slides smoothly over 1 second — natural!

 **Key Rule:** Put the **transition** property in the NORMAL style, not inside **:hover**!

# Duration: How Fast or Slow?



0.2s–0.3s

Very Fast

Quick blink

✓ Micro-interactions

0.5s

Fast

Snappy & clean

✓ Button colour changes

1s

Medium

Comfortable

✓ Most hover effects

2s–3s

Slow

Dramatic sunrise

✓ Loading, special reveals

5s+

Very Slow

Dreamy

✓ Background colour shifts

# Properties You Can Transition



`background-color`

**Fill colour**

*orange → green*



`font-size`

**Text size**

*16px → 24px*



`transform: scale()`

**Grow / shrink**

*scale(1) → scale(1.3)*



`color`

**Text colour**

*black → white*



`border-radius`

**Corner shape**

*0px → 50%*



`transform: rotate()`

**Spin**

*0deg → 45deg*



`width / height`

**Element size**

*100px → 200px*



`opacity`

**Transparency**

*1 → 0.5*



`box-shadow`

**Drop shadow**

*none → 0 4px 20px ...*





# Hands-On Activity

## MUST DO

### Colour Change on Hover

Create a coloured box that smoothly changes background colour on hover. Use `transition: background-color 1s;` in `.box` style.

## SHOULD DO

### Try Three Speeds

Change duration to 0.3s, 1s, and 3s one at a time. Notice how each feels different. Which feels most natural?

## STRETCH

### Grow on Hover

Add `transform: scale(1.3);` in `.box:hover`. Transition both colour AND transform together with commas!

## ★ Starter Code

```
<style>
.box {
  width: 200px;
  height: 120px;
  background-color: orange;
  border-radius: 12px;
  /* 📍 ADD HERE */
  transition: bg-color 1s;
}

.box:hover {
  background-color: green;
  /* scale(1.3); */
}

</style>
```



# Common Mistakes to Avoid



## Transition inside :hover

Putting transition: inside the :hover block — it won't animate back to original!

✓ **Fix:** Always put transition in the NORMAL style block.



## Missing 's' after number

Writing "1" instead of "1s" — the browser ignores it!

✓ **Fix:** Always write 1s, 0.3s, 2s — never just a plain number.



## Non-animatable properties

Trying to transition display or font-family — these cannot be animated.

✓ **Fix:** Stick to: color, transform, opacity, width, height, border-radius.



## Property not changing

Using transition on a property that isn't actually different in :hover — nothing happens!

✓ **Fix:** Make sure the value actually CHANGES between normal and :hover.



# Key Terms Recap

transition

**CSS property that enables smooth change**

*transition: background-color 1s;*

property

**What to change smoothly**

*background-color, transform, opacity*

duration

**How long the change takes**

*0.5s (fast) / 1s (medium) / 3s (slow)*

:hover

**Trigger — activates on mouse hover**

*.box:hover { background-color: green; }*

selector

**Which HTML element to style**

*.box { ... }*

transform: scale()

**Resize the element**

*scale(1.3) makes it 30% bigger*



# Homework Challenge

1

MUST DO

## Button with Colour + Size Change

Create a `<button>` that changes background colour AND grows (`transform: scale(1.1)`) on hover. Use transition for both. Try from memory first!

2

BONUS

## Square to Circle

A square div (`border-radius: 0%`). On hover → circle (`border-radius: 50%`). Smooth transition + colour change at the same time!

3

SHOW OFF 🏆

## Teach Your Family!

Show your webpage to your family. Explain CSS Transition in your own words. What does `:hover` do? Teaching = the best way to remember!

🔍 Bonus: Look up transition-timing-function — try *ease*, *ease-in*, *ease-out*, *linear*. Which do you like best?



# You're a CSS Animator Now!

*Every button glow, every smooth hover — you now know how it's made!*

★ Stars twinkling in a night sky   🪔 Diwali diya flame flickering   ⌚ A loading spinner   ✨ Elements that fade in on page load

## Session Takeaways

- ✓ CSS Transition makes changes smooth — no instant jumps
- ✓ transition property goes in the NORMAL style block
- ✓ :hover is the trigger — fires on mouse over
- ✓ Duration controls speed — try 0.3s, 1s, 3s
- ✓ You can transition multiple properties with commas